

```
""bash echo -e "Hello, I'm a language
model" | transformers run --task text-
generation --model openai-
community/gpt2 --device 0 ""
```

https://huggingface.co/docs/transformers/main/en/model_doc/gpt2#transformers

Detailed Documentation

class

inline">transformers.GPT2Configtransformers.GPT2Confighttps<

class

`transformers.GPT2Configtransformers.GPT2Confighttps` ://github.com/huggir
"vocab_size", "val": " = 50257"}, {"name": "n_positions", "val": " = 1024"},
{"name": "n_embd", "val": " = 768"}, {"name": "n_layer", "val": " = 12"}, {"name":
"n_head", "val": " = 12"}, {"name": "n_inner", "val": " = None"}, {"name":
"activation_function", "val": " = 'gelu_new'"}, {"name": "resid_pdrop", "val": " =
0.1"}, {"name": "embd_pdrop", "val": " = 0.1"}, {"name": "attn_pdrop", "val": " =
0.1"}, {"name": "layer_norm_epsilon", "val": " = 1e-05"}, {"name":
"initializer_range", "val": " = 0.02"}, {"name": "summary_type", "val": " =
'cls_index'"}, {"name": "summary_use_proj", "val": " = True"}, {"name":

```
"summary_activation", "val": " = None"}, {"name": "summary_proj_to_labels",  
"val": " = True"}, {"name": "summary_first_dropout", "val": " = 0.1"}, {"name":  
"scale_attn_weights", "val": " = True"}, {"name": "use_cache", "val": " = True"},  
{"name": "bos_token_id", "val": " = 50256"}, {"name": "eos_token_id", "val": " =  
50256"}, {"name": "scale_attn_by_inverse_layer_idx", "val": " = False"}, {"name":  
"reorder_and_upcast_attn", "val": " = False"}, {"name": "***kwargs", "val": ""}]-  
**vocab_size** ( int, *optional*, defaults to 50257) --
```

Vocabulary size of the GPT-2 model. Defines the number of different tokens that can be represented by the

`inputs_ids` passed when calling `[GPT2Model]`
(/docs/transformers/main/en/model_doc/gpt2#transformers.GPT2Model) or
`TFGPT2Model`.

- `**n_positions**` (`int`, *optional*, defaults to 1024) --

The maximum sequence length that this model might ever be used with. Typically set this to something large

just in `case` (e.g., 512 or 1024 or 2048).

- `**n_embd**` (`int`, *optional*, defaults to 768) --

Dimensionality of the embeddings and hidden states.

- `**n_layer**` (`int`, *optional*, defaults to 12) --

Number of hidden layers in the Transformer encoder.

`class` `<span class="code-`

`inline">transformers.GPT2Tokenizertransformers.GPT2Tokenizer`

`class`

```
transformers.GPT2Tokenizertransformers.GPT2Tokenizerhttps://github.com,  
"vocab_file", "val": "", {"name": "merges_file", "val": ""}, {"name": "errors", "val":  
" = 'replace'"}, {"name": "unk_token", "val": " = '<|endoftext|>'"}, {"name":  
"bos_token", "val": " = '<|endoftext|>'"}, {"name": "eos_token", "val": " =  
'<|endoftext|>'"}, {"name": "pad_token", "val": " = None"}, {"name":  
"add_prefix_space", "val": " = False"}, {"name": "add_bos_token", "val": " =  
False"}, {"name": "**kwargs", "val": ""}- **vocab_file** ( str ) --
```

Path to the vocabulary file.

- `**merges_file**` (`str`) --

Path to the merges file.

- `**errors**` (`str` , *optional*, defaults to `"replace"`) --

Paradigm to follow when decoding bytes to UTF-8. See

[bytes.decode](https://docs.python.org/3/library/stdtypes.html#bytes.decode) for more information.

- `**unk_token**` (`str` , *optional*, defaults to `"<|endoftext|>"`) --

The unknown token. A token that is not in the vocabulary cannot be converted to an ID and is set to be this

- `**bos_token**` (`str`, *optional*, defaults to `"<|endoftext|>"`) --

save_vocabularytransformers.GPT2Tokenizer.save_vocabularyht

save_vocabularytransformers.GPT2Tokenizer.save_vocabularyhttps://github.com,
 "save_directory", "val": ": str"}, {"name": "filename_prefix", "val": ":
 typing.Optional[str] = None"]}]

class <span class="code-

inline">transformers.GPT2TokenizerFasttransformers.GPT2Toker

class

`transformers.GPT2TokenizerFasttransformers.GPT2TokenizerFasthttps` ://git
 "vocab_file", "val": " = None"}, {"name": "merges_file", "val": " = None"},
 {"name": "tokenizer_file", "val": " = None"}, {"name": "unk_token", "val": " =
 '<|endoftext|>'"}, {"name": "bos_token", "val": " = '<|endoftext|>'"}, {"name":
 "eos_token", "val": " = '<|endoftext|>'"}, {"name": "add_prefix_space", "val": " =
 False"}, {"name": "**kwargs", "val": ""}] - `**vocab_file**` (`str`, *optional*) --

Path to the vocabulary file.

- `**merges_file**` (`str`, *optional*) --

Path to the merges file.

- `**tokenizer_file**` (`str`, *optional*) --

Path to [tokenizers](https://github.com/huggingface/tokenizers) `file` (generally has a .json extension) that

contains everything needed to load the tokenizer.

- `**unk_token**` (`str`, *optional*, defaults to `"<|endoftext|>"`) --

The unknown token. A token that is not in the vocabulary cannot be converted to an ID and is set to be this

- `**bos_token**` (`str`, *optional*, defaults to `"<|endoftext|>"`) --

`class` `<span class="code-`

`inline">transformers.models.gpt2.modeling_gpt2.GPT2DoubleHe`

`class`

```
transformers.models.gpt2.modeling_gpt2.GPT2DoubleHeadsModelOutputtrans
"loss", "val": ": typing.Optional[torch.FloatTensor] = None"}, {"name": "mc_loss",
"val": ": typing.Optional[torch.FloatTensor] = None"}, {"name": "logits", "val": ":
typing.Optional[torch.FloatTensor] = None"}, {"name": "mc_logits", "val": ":
typing.Optional[torch.FloatTensor] = None"}, {"name": "past_key_values", "val":
": typing.Optional[ transformers.cache_utils.Cache ] = None"}, {"name":
"hidden_states", "val": ": typing.Optional[tuple[torch.FloatTensor]] = None"},
{"name": "attentions", "val": ": typing.Optional[tuple[torch.FloatTensor]] =
None"]}- loss ( torch.FloatTensor of shape (1,), *optional*, returned
when labels is provided) --
```

Language modeling loss.

- **mc_loss** (`torch.FloatTensor` of shape `(1,)`, *optional*, returned when `mc_labels` is provided) --

Multiple choice classification loss.

- **logits** (`torch.FloatTensor` of shape `(batch_size, num_choices, sequence_length, config.vocab_size)`) --

Prediction scores of the language modeling `head` (scores for each vocabulary token before SoftMax).

- **mc_logits** (`torch.FloatTensor` of shape `(batch_size, num_choices)`) --

Prediction scores of the multiple choice classification `head` (scores for each choice before SoftMax).

- **past_key_values** (`Cache`, *optional*, returned when `use_cache=True` is passed or when `config.use_cache=True`) --

It is a `[Cache]` (/docs/transformers/main/en/internal/generation_utils#transformers.Cache) instance. For more details, see our [\[kv cache guide\]](https://huggingface.co/docs/transformers/en/kv_cache) (https://huggingface.co/docs/transformers/en/kv_cache).

class `<span class="code-`

`inline">transformers.GPT2Modeltransformers.GPT2Modelhttps</`

class

```
transformers.GPT2Model transformers.GPT2Model https://github.com/huggingf
"config", "val": ""}] - **config** ([GPT2Model]
(/docs/transformers/main/en/model_doc/gpt2# transformers.GPT2Model )) --
```

Model configuration class with all the parameters of the model. Initializing with a config file does not

load the weights associated with the model, only the configuration. Check out the

```
[ from_pretrained ()]
(/docs/transformers/main/en/main_classes/model# transformers.PreTrainedMoc
method to load the model weights.0
```

The bare Gpt2 Model outputting raw hidden-states without any specific head on top.

This model inherits from [PreTrainedModel] (/docs/transformers/main/en/main_classes/model# transformers.PreTrainedMoc) Check the superclass documentation for the generic methods the

library implements for all its `model` (such as downloading or saving, resizing the input embeddings, pruning heads

This model is also a PyTorch [torch.nn.Module] (<https://pytorch.org/docs/stable/nn.html#torch.nn.Module>) subclass.

Use it as a regular PyTorch Module and refer to the PyTorch documentation for all matter related to general usage

```
forwardtransformers.GPT2Model.forwardhttps://github.com/huggingface/transfor
"input_ids", "val": ": typing.Optional[torch.LongTensor] = None"}, {"name":
"past_key_values", "val": ": typing.Optional[ transformers.cache_utils.Cache ]
= None"}, {"name": "cache_position", "val": ": typing.Optional[torch.LongTensor]
= None"}, {"name": "attention_mask", "val": ": typing.Optional[torch.FloatTensor]
= None"}, {"name": "token_type_ids", "val": ": typing.Optional[torch.LongTensor]
= None"}, {"name": "position_ids", "val": ": typing.Optional[torch.LongTensor] =
None"}, {"name": "inputs_embeds", "val": ": typing.Optional[torch.FloatTensor] =
None"}, {"name": "encoder_hidden_states", "val": ":
typing.Optional[torch.Tensor] = None"}, {"name": "encoder_attention_mask",
"val": ": typing.Optional[torch.FloatTensor] = None"}, {"name": "use_cache",
"val": ": typing.Optional[bool] = None"}, {"name": "output_attentions", "val": ":
typing.Optional[bool] = None"}, {"name": "output_hidden_states", "val": ":
typing.Optional[bool] = None"}, {"name": "return_dict", "val": ":
typing.Optional[bool] = None"}, {"name": "**kwargs", "val": ""}]- **input_ids**
( torch.LongTensor of shape (batch_size, input_ids_length) ) --
```

forwardtransformers.GPT2Model.forwardhttps://github.com/hugg

```
forwardtransformers.GPT2Model.forwardhttps://github.com/huggingface/transfor
"input_ids", "val": ": typing.Optional[torch.LongTensor] = None"}, {"name":
"past_key_values", "val": ": typing.Optional[ transformers.cache_utils.Cache ]
= None"}, {"name": "cache_position", "val": ": typing.Optional[torch.LongTensor]
= None"}, {"name": "attention_mask", "val": ": typing.Optional[torch.FloatTensor]
= None"}, {"name": "token_type_ids", "val": ": typing.Optional[torch.LongTensor]
= None"}, {"name": "position_ids", "val": ": typing.Optional[torch.LongTensor] =
None"}, {"name": "inputs_embeds", "val": ": typing.Optional[torch.FloatTensor] =
```



```
None"}, {"name": "encoder_hidden_states", "val": ":
typing.Optional[torch.Tensor] = None"}, {"name": "encoder_attention_mask",
"val": ": typing.Optional[torch.FloatTensor] = None"}, {"name": "use_cache",
"val": ": typing.Optional[bool] = None"}, {"name": "output_attentions", "val": ":
typing.Optional[bool] = None"}, {"name": "output_hidden_states", "val": ":
typing.Optional[bool] = None"}, {"name": "return_dict", "val": ":
typing.Optional[bool] = None"}, {"name": "**kwargs", "val": ""}] - **input_ids**
( torch.LongTensor of shape (batch_size, input_ids_length) ) --
```

```
input_ids_length = sequence_length if past_key_values is None else
past_key_values.get_seq_length() ( sequence_length of input past key
value states). Indices of input
```

sequence tokens in the vocabulary.

If `past_key_values` is used, only `input_ids` that do not have their past
calculated should be passed as

Indices can be obtained using `[AutoTokenizer]`
(/docs/transformers/main/en/model_doc/auto#transformers.AutoTokenizer).

See `[PreTrainedTokenizer.encode()]`
(/docs/transformers/main/en/internal/tokenization_utils#transformers.PreTrainedTokenizer.encode)
and

`[PreTrainedTokenizer.__call__()]`
(/docs/transformers/main/en/internal/tokenization_utils#transformers.PreTrainedTokenizer.__call__)
for details.

[What are input IDs?]([./glossary#input-ids](/glossary#input-ids))

- `**past_key_values**` (`~cache_utils.Cache`, *optional*) --

Pre-computed hidden-`states` (key and values in the self-attention blocks and in the cross-attention)

class `<span class="code-`

inline">transformers.GPT2LMHeadModeltransformers.GPT2LMH
